

HAMRA, Sweden

WMO: 023290

Lat: 61.6599N

Lon: 14.9918E

Elev: 455

StdP: 95.98

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-16.4	-21.4	0.6	-17.9	-19.2	0.7	-15.4	7.8	0.2	7.0	-0.2	2.2	350	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	24.5	15.9	22.6	15.2	20.7	14.4	17.4	21.8	16.4	20.5	15.4	19.2	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.8	18.5	14.7	11.0	17.8	13.7	10.3	16.9	50.4	21.8	47.5	20.4	44.5	19.1	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.5	5.0	DB	-21.5	26.9	2.5	1.3	-23.3	27.9	-24.7	28.6	-26.1	29.4	-27.9	30.4	(4)
(5)			WB	-21.5	18.4	2.3	1.1	-23.2	19.2	-24.6	19.9	-25.9	20.5	-27.6	21.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-5.9	-5.4	-2.0	3.1	8.3	12.3	14.9	13.4	9.1	3.4	-1.4	-4.3	(6)	
	DBStd	8.38	4.95	5.24	4.49	3.77	4.10	3.49	3.17	3.06	3.18	3.76	4.20	5.15	(7)	
	HDD10.0	2653	492	430	371	208	85	14	1	6	53	206	343	444	(8)	
	HDD18.3	5300	750	663	629	456	312	184	115	156	276	463	593	703	(9)	
	CDD10.0	409	0	0	0	2	31	83	154	111	27	2	0	0	(10)	
	CDD18.3	15	0	0	0	0	0	3	9	3	0	0	0	0	(11)	
	CDH23.3	109	0	0	0	0	3	24	69	13	0	0	0	0	(12)	
	CDH26.7	9	0	0	0	0	0	1	8	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.4	2.5	2.6	2.8	2.6	2.5	2.3	2.1	2.0	2.2	2.5	2.6	2.5	(14)	
(15) Precipitation	PrecAvg	716	44	31	32	44	54	81	96	95	69	67	59	45	(15)	
	PrecMax	1070	99	69	72	149	111	152	242	168	151	173	174	76	(16)	
	PrecMin	521	7	6	2	3	9	15	12	46	18	13	18	4	(17)	
	PrecStd	119	24	14	17	35	25	30	54	35	36	39	35	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	6.3	10.7	17.1	23.7	26.0	27.8	25.0	19.7	14.3	9.3	6.5	(19)	
		MCWB	2.8	2.7	5.0	9.0	13.4	15.1	17.0	17.2	13.5	10.9	8.0	4.7	(20)	
	2%	DB	3.3	4.5	7.9	14.6	20.9	23.6	25.4	22.5	17.6	11.7	7.1	4.7	(21)	
		MCWB	1.8	2.0	3.9	7.7	12.2	14.5	16.9	16.3	12.8	9.3	6.1	3.2	(22)	
	5%	DB	2.0	2.9	5.9	12.0	18.3	21.4	23.2	20.6	15.7	10.1	5.4	3.4	(23)	
		MCWB	0.9	1.1	2.5	6.2	11.2	13.8	15.9	15.2	12.0	8.1	4.6	2.1	(24)	
	10%	DB	0.7	1.4	4.2	9.5	15.6	18.8	21.0	18.7	14.0	8.7	4.0	2.1	(25)	
		MCWB	-0.4	0.0	1.5	4.8	9.8	12.5	15.2	14.4	11.1	7.1	3.4	1.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.6	5.9	10.4	14.7	17.6	18.7	18.2	14.6	11.6	8.4	5.2	(27)	
		MCDB	4.4	5.4	9.7	16.2	20.6	22.6	24.4	22.2	17.7	13.2	9.1	6.0	(28)	
	2%	WB	2.0	2.3	4.2	8.2	13.2	15.3	17.6	17.1	13.5	9.8	6.2	3.5	(29)	
		MCDB	3.0	4.0	7.2	13.6	19.4	20.8	22.5	20.9	16.6	11.2	6.9	4.5	(30)	
	5%	WB	1.0	1.2	2.8	6.6	11.7	14.2	16.8	16.1	12.5	8.6	4.7	2.2	(31)	
		MCDB	2.0	2.8	5.4	11.2	16.8	19.7	21.2	19.5	15.1	9.8	5.2	3.2	(32)	
	10%	WB	-0.3	0.1	1.7	5.3	10.3	13.2	15.9	15.0	11.5	7.4	3.4	1.0	(33)	
		MCDB	0.5	1.3	3.9	8.9	15.1	18.3	20.1	17.9	13.6	8.5	3.9	1.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.2	6.5	7.7	8.7	9.0	8.5	7.8	6.9	5.3	4.0	4.5	(35)	
		MCDBR	5.1	5.7	8.0	11.2	12.6	12.4	11.4	10.1	9.0	7.0	5.3	5.7	(36)	
	5% WB	MCWBR	4.6	4.3	5.4	6.2	6.6	6.0	5.1	5.0	5.3	4.9	4.8	5.3	(37)	
		MCWBR	4.9	5.6	7.6	10.4	11.1	10.9	9.8	8.7	8.4	6.1	5.0	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.222	0.273	0.299	0.333	0.338	0.333	0.349	0.341	0.310	0.296	0.240	0.341	(40)	
		taud	2.152	2.294	2.361	2.413	2.462	2.500	2.465	2.480	2.556	2.490	2.222	2.766	(41)	
	Edn at Noon		513	714	825	862	885	895	868	846	817	692	480	355	(42)	
		Edh at Noon	34	60	80	94	97	95	96	87	67	51	31	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.26	0.93	2.52	4.03	4.69	5.31	4.85	3.71	2.40	1.10	0.33	0.14	(44)	
	RadStd		0.04	0.16	0.24	0.40	0.50	0.41	0.59	0.33	0.32	0.18	0.08	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page